

Monday Reading Assessment: Unit 3, Magnetic Forces and Fields

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1 Memory Bank

- $\vec{F} = q\vec{v} \times \vec{B}$... The Lorentz force, one version.
- $\vec{F} = i\vec{L} \times \vec{B}$... The Lorentz force, another version.
- $\vec{\tau} = \vec{r} \times \vec{F}$... Definition of torque.
- $\vec{\tau} = \vec{\mu} \times \vec{B}$... Torque on a current carrying loop in a uniform B-field.
- $\vec{\mu} = Ni\vec{A}$... Definition of the magnetic moment.

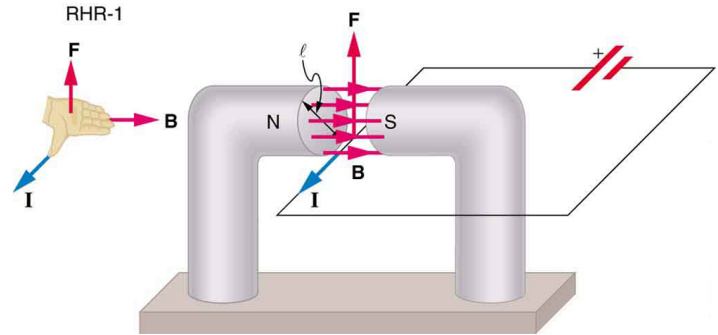


Figure 1: A permanent magnet creates a magnetic field to the *right*, while a voltage pushes a current *out of the page*. The force is measured to occur in the *upward* direction.

2 Magnetic Force on a Wire

1. Consider Fig. 1, in which a current passes through a *magnetic field* generated by a permanent magnet. (a) If the length of wire in the B-field is $L = 10$ cm, the magnetic field is $B = 10^{-1}$ Tesla, the voltage is $V = 24$ Volts, and the effective resistance in the wire is $R = 2\Omega$, what is the force upwards on the wire? (b) If the wire was attached to a scale, and the scale has a mass of $m = 24$ grams on it, what would the scale read if we turn on the current?¹

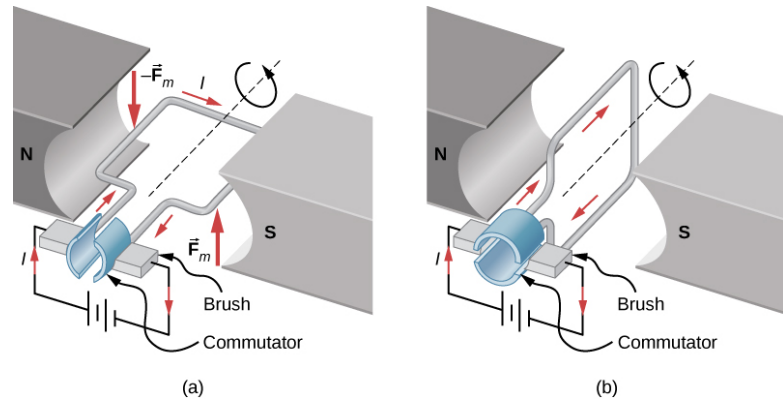


Figure 2: An illustration of how a power generator works. This version uses DC current and a commutator.

3 Torque, revisited

1. Suppose we apply a force $\vec{F} = 10\hat{j}$ N (upward) on the end of a 10 cm wrench it as *loosens* a bolt at the origin. The bolt is to the left of the wrench. What is the magnitude *and direction* of the torque?

and the circuit resistance is 50Ω . Also assume that there is just one loop of wire in the rotor. What is the *maximum torque* the system could achieve?

4 The Power Generator

1. Consider Fig. 2, in which a DC power generator is depicted inside a 0.05 T B-field. Suppose the area of the loop is 10^{-2} m², the voltage in the circuit is 24 V,

2. What would the maximum torque be if there were $N = 100$ turns of wire?

¹Hint: it won't say 24 grams.